Exchange Properties

Thm: $M=(E,\mathcal{I})$. Let B_1, B_2 be bases

Take $x \in B_1 \setminus B_2$. Then $\exists\, y \in B_2 \setminus B_1$

st: $B_1 - x + y$, $B_2 - y + x$ are both

bases.

pf: $B_2 + x$ has a unique circuit - C

$x \in \text{span}((C-x))$ (since $\text{rank}((C)) = \text{rank}((C-x)).$)

$\Rightarrow x \in \text{span}(B_1 \cup C - x) \Rightarrow \text{span}(B_1 \cup C - x) = \text{span}(\underline{B_1 \cup C}) = E$

$\Rightarrow \exists\, B_3$ with $B_1 - x \subseteq B_3 \subseteq B_1 \cup C - x$

Moreover, C is the only circuit in $B_2 + x$

$\Rightarrow \underline{B_2 + x - y}$ has no circuit $\Rightarrow$ it is a base.

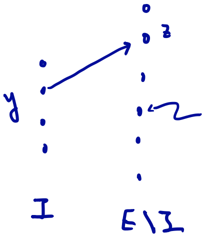
A Bipartite Directed Graph

$$M = (E, \mathcal{I})$$

For $I \in \mathcal{I}$, define a directed bipartite graph: $D_M(I) = (E, A(I))$

where

$$A(I) = \{ (y,z) : y \in I,\ z \in E \setminus I,\ I - y + z \in \mathcal{I} \}$$



$I - y + z \in \mathcal{I}$

if this element is in $\mathrm{span}(I - y)$ then there is no edge from y.

A(I) and Perfect Matchings

Corollary: Let $I, J \in \mathcal{I}$ with $|I| = |J| = k$

Then, $A(I)$ contains a perfect matching on $I \Delta J$

pf: Let M' be the truncated matroid: $M' = (E, \mathcal{I}')$ where $\mathcal{I}' = \{I \in \mathcal{I} : |I| \leq k\}$. Now I, J are bases in M'.

We prove Corollary by induction $|I \setminus J| \geq 1$.

Pick $y \in I \setminus J$. By our exchange theorem, $\exists z \in J \setminus I$ s.t.

$I - y + z \in \mathcal{I}'$, $\underline{J - z + y} \in \mathcal{I}'$.

Let $J' = J - z + y$. By induction, $A(I)$ has a perfect matching, N, on $I \Delta J'$.

Then $N \cup \{(y, z)\}$ is a perfect matching on $I \Delta J$. ∎

The Result in Pictures